

MOHAMMAD NAVED

FOUNDER & LEAD AI ENGINEER · CODTREX AI | FULL-STACK PRODUCT ARCHITECT

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PROFILE SUMMARY

Founder & Lead AI Engineer with a track record of shipping production-grade, full-stack products — from Google-certified IoT hardware ecosystems to B2B SaaS platforms and AI-powered studio products. Founded Codtrex AI, a boutique AI engineering studio delivering custom LLM, IoT, and cloud systems for clients. Strong across the entire stack: embedded firmware (ESP32), cloud infrastructure (AWS, Docker), modern web frontends (React 19, Next.js), and AI integration (Gemini, multi-LLM). Actively seeking engineering roles and consulting engagements where ownership and zero-to-one velocity are valued.

CORE EXPERTISE

- **AI/LLM Engineering** — production Gemini API integration, multi-LLM orchestration, agentic workflows, prompt architecture, RAG systems
- **Full-Stack Product Development** — React 19, Next.js 15/16, TypeScript, Tailwind v4, Vite, Supabase (PostgreSQL), FastAPI
- **IoT & Embedded Systems** — ESP32 firmware (Arduino/C++), MQTT broker (AWS IoT Core), mTLS device auth, AWS App Runner, Docker, OTA updates
- **Cloud & DevOps** — AWS (IoT Core, App Runner, ECR, S3), Vercel, Supabase, Docker, GitHub Actions CI
- **Mobile** — React Native (Expo), Capacitor Android (Google Play shipping)
- **Studio & Client Delivery** — end-to-end product ownership from spec to deployment; Google Smart Home certification process

VENTURES & EXPERIENCE

Codtrex AI — Boutique AI Engineering Studio

2025 – Present

Founder & Lead Engineer | codtrex.vercel.app | React 19, Vite, Tailwind v4, Supabase, Gemini API, Vercel

- Founded and operate a boutique AI engineering studio offering end-to-end LLM integration, full-stack web products, IoT systems, and technical consultation to clients
- Built and shipped the studio's public website with AI chat assistant, service tiers, and case studies — live at codtrex.vercel.app
- Designed multi-LLM product architecture and client-facing AI solutions across web, mobile, and embedded platforms

4Layers — Google Certified Smart Home IoT Ecosystem

2025 – Present

Founder & Systems Engineer | ESP32, AWS IoT Core, MQTT, mTLS, React Native, FastAPI, Docker, AWS App Runner

- Achieved **official Google Smart Home certification** — passed 71+/75 tests in the Google Home Test Suite; devices are listed and functional in the Google Home ecosystem
- Architected a **3-source real-time state sync** (ESP32 hardware ↔ AWS IoT Core MQTT ↔ Google Home) ensuring zero state divergence across physical buttons, app controls, and voice commands
- Engineered QA Hold Mode to prevent Google Home from reverting local hardware states during test execution — a critical certification-blocking fix discovered and resolved independently
- Built OTA 2.0 with version pinning and rollback safety — firmware update pipeline without physical access to deployed devices
- Designed custom ESP32 hardware nodes: 4-channel relay switch boards + Triac-based multi-speed fan regulator; mTLS certificates provisioned per-device via AWS IoT Core

MediStock — Cloud Pharmacy Management SaaS

2025 – Present

Founder & Product Architect | React 18, Supabase, PostgreSQL, Capacitor Android, Vercel | medistock-pharma.vercel.app

- Built and shipped a full B2B pharmacy SaaS: inventory management, expiry tracking, billing, and supplier management for pharmacies
- Deployed cross-platform: live web app + Android APK via Capacitor — targeting Google Play submission
- Designed multi-tenant Supabase (PostgreSQL) schema with Row Level Security for pharmacy data isolation

PROJECTS

UsedTech Market — C2C Electronics Marketplace2025

Product Architect | Next.js 15/16, React 19, TypeScript, Supabase, Tailwind v4

- Architected a consumer-to-consumer marketplace for buying and selling used electronics with real-time listings, search filters, and seller dashboards
- Built on Next.js 15/16 App Router with SSR for SEO-optimized product pages and Supabase real-time subscriptions for live listing updates

AutoApply AI — Automated Job Application System2025

Self-Initiated | AI Automation | Python, Gemini AI, Multi-LLM

- Designed a multi-step prompt chain that reads job descriptions, extracts requirements, and generates tailored resumes and cover letters via structured LLM calls
- Engineered retry and fallback prompt logic to handle malformed or incomplete model outputs in production

DataLens AI — Full-Stack AI Analytics Platform2025

Self-Initiated | React, TypeScript, Recharts, Gemini AI, DuckDB, SQL

- Built an AI analytics platform with a natural-language-to-SQL chat interface and ML Lab module for model output explanation and insight surfacing
- Designed prompt templates for consistent LLM output across varying data schemas and edge cases

Interactive Portfolio & PDF Automation Pipeline2025

Self-Initiated | React 19, TypeScript, Framer Motion, Tailwind v4, Node.js, Puppeteer, Gemini AI

- Built a single-page interactive portfolio featuring hardware-accelerated animations, an AI chatbot (Gemini-powered), and a context-aware navigation system
- Automated resume PDF regeneration via a Node.js/Puppeteer script — pixel-perfect A4 export from live HTML source, eliminating manual export steps

TECHNICAL SKILLS & COMPETENCIES

- **AI / LLM:** Gemini API, Multi-LLM Orchestration, Prompt Engineering, Agentic Workflows, RAG Systems
- **Frontend:** React 19, Next.js 15/16, TypeScript, Tailwind v4, Vite, Framer Motion
- **Backend:** FastAPI (Python), Node.js, REST APIs, Supabase (PostgreSQL), SQL
- **IoT / Embedded:** ESP32 (Arduino/C++), MQTT, AWS IoT Core, mTLS, OTA Firmware Updates, Triac control
- **Cloud & DevOps:** AWS (IoT Core, App Runner, ECR, S3), Docker, Vercel, GitHub Actions
- **Mobile:** React Native (Expo), Capacitor Android
- **Languages:** TypeScript, Python, C++ (Arduino), SQL, JavaScript

EDUCATION

B.Tech, Computer Science — RGPV University, Indore2025

CERTIFICATIONS & ACHIEVEMENTS

- Google Smart Home Certification — 4Layers IoT Ecosystem (71+/75 tests passed, Google Home Test Suite)
- Generative AI with Gemini — Prompt Engineering & LLM Application Development
- Python for Data Analysis & Visualization — Self-directed
- SQL for Data Analytics — Relational Database Workflows
- Power BI — Interactive Dashboard Design & Reporting